DS5003A Data Engineering

Assignment-1, Date: 31-July-2026

Timing: 2:00 PM to 4:45 PM **Not Graded**

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1. Write a Python program that takes a temperature in Celsius as input and converts it into Fahrenheit.

   Use the following formula:

   $$F = \frac{9}{5}C + 32$$

   Example:

   ```
   Enter temperature in Celsius: 25
   Temperature in Fahrenheit: 77.0
   ```

2. Write a Python program to calculate the simple interest and total amount. Take the principal amount, rate of interest, and time in years as input.

   $$\text{Simple Interest} = \frac{P \times R \times T}{100}$$

   $$\text{Total Amount} = P + \text{Simple Interest}$$

3. Write a Python program that takes three numbers as input and prints the largest number. Do not use the built-in `max()` function.

   Example:

   ```
   Enter the first number: 14
   Enter the second number: 27
   Enter the third number: 19
   Largest number: 27
   ```

4. Write a Python program that takes the lengths of three sides as input. Check whether the given sides can form a valid triangle. A triangle is valid if the sum of any two sides is greater than the third side.

5. Write a Python program that takes a positive integer as input, reverses its digits using a loop, and checks whether it is a palindrome.

   A palindrome number remains the same when its digits are reversed.

   Hint: Use the modulo operator `%` to obtain the last digit of the number and the floor division operator `//` to remove the last digit.

   Example:

```
Enter a number: 1221
Reversed number: 1221
The number is a palindrome.
```

6. Write a Python program to calculate the sum of the digits of a given positive integer using a `while` loop.

   Hint: Use `number % 10` to obtain the last digit and `number // 10` to remove the last digit.

   Example:

   ```
   Enter a number: 5832
   Sum of digits: 18
   ```

7. Write a Python program that takes a person's name and age as input and displays them in a single sentence. Convert the age into an integer, increase it by 1, and then explicitly convert it back into a string before concatenating it with the other strings.

   The program should demonstrate the use of:

   - `int()` to convert a string into an integer
   - `str()` to convert an integer into a string
   - The `+` operator for string concatenation

   Example:

   ```
   Enter your name: Arun
   Enter your age: 20
   Hello Arun, you will be 21 years old next year.
   ```

8. Write a Python program that takes two positive integers as input and finds their greatest common divisor. Do not use functions from the `math` module.

   Example:

   ```
   Enter the first number: 36
   Enter the second number: 48
   GCD: 12
   ```

9. Write a Python program that takes a number n and a limit m as input and prints the multiplication table of n from 1 to m.

   Example:

```
Enter the number: 7
Enter the limit: 5

7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
```

10. Write a Python program to print the following pattern for a given number of rows n. The value of n must be taken as input from the user.

    For $n = 5$:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

11. Write a Python program to print a centred star pyramid of height H, where $H \geq 2$. The height must be taken as input from the user. Use nested loops to construct the pattern.

    For $H = 5$:

```
    *
   ***
  *****
 *******
*********
```

12. Write a Python program to read n integers from the user and store them in a list. Display the elements of the list, calculate their sum, and find the largest element.

    Do not use the built-in `sum()` or `max()` functions.

    Example:

```
Enter the number of elements: 5
Enter element 1: 12
Enter element 2: 7
Enter element 3: 25
Enter element 4: 10
Enter element 5: 16

List elements: [12, 7, 25, 10, 16]
```

```
Sum of elements: 70
Largest element: 25
```

13. Define a function `is_perfect(n)` that returns `True` if n is a perfect number and `False` otherwise.

    A perfect number is a positive integer that is equal to the sum of its proper divisors, excluding the number itself. Use the function to display all perfect numbers between 1 and a limit entered by the user.

    Example:

    ```
    Enter the upper limit: 1000
    Perfect numbers between 1 and 1000: 6 28 496
    ```

    For example, 28 is a perfect number because:

    $$1 + 2 + 4 + 7 + 14 = 28$$